

From Adam to AMSGrad: Exact Counterexample Replication, Diagnostic Dynamics, and Phase-Boundary Scaling

Research Report

Adaptive Gradient Methods Replication & Extension Lineage

<https://github.com/rayyan/AMS-grad>

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Abstract

Adaptive gradient algorithms, led by ADAM, are fundamental to deep learning optimization. However, Reddi et al. [4] revealed a fundamental flaw in ADAM’s convergence proof under the Online Convex Optimization framework: when informative gradients occur sparsely, exponential moving averages allow the second-moment estimate to decay, violating the positive semi-definiteness condition of the metric matrix ($\Gamma_t \not\succeq 0$) and causing divergence to sub-optimal boundaries. To resolve this, AMSGRAD enforces monotonic non-decrease of second-moment estimates via a running maximum ($\hat{v}_t = \max(\hat{v}_{t-1}, v_t)$). In this work, we present a complete, from-scratch NumPy reproduction and extension of the ADAM $\rightarrow$ AMSGRAD lineage. We faithfully reconstruct both deterministic (Theorem 3) and stochastic (Section 3) counterexamples, introducing real-time Γ_t tracking that visualizes the theoretical breakdown. Furthermore, we execute a pre-registered extension evaluating the (β_2, C) phase boundary across 4,500 stochastic runs ($N = 100$ seeds per cell). Our empirical results refute the simple C^{-1} scaling law (observed OLS slope -3.78 ± 1.60), demonstrating that burst sparsity compounds ADAM’s vulnerability super-linearly as consecutive negative steps drain the first moment. Finally, we discuss why AMSGRAD’s historical max locking induces practical conservatism in non-stationary deep learning regimes.

1 Introduction

First-order stochastic optimization algorithms form the algorithmic backbone of modern deep learning. Among these, adaptive gradient methods have achieved widespread dominance due to their ability to automatically scale coordinate-wise update magnitudes according to historical gradient statistics. The seminal work of Kingma and Ba [3] introduced ADAM, which synthesizes the advantages of ADAGRAD [1]—coordinate-wise scaling suitable for sparse gradients—and RMSPROP [5]—exponential moving averages capable of adapting to non-stationary

objective landscapes. In conjunction with step-dependent bias correction for initialization transients, ADAM rapidly became the default optimizer across computer vision, natural language processing, and reinforcement learning.

Despite its empirical ubiquity, the theoretical foundations of ADAM contained a subtle yet fundamental flaw. In their breakthrough analysis, Reddi et al. [4] demonstrated that the convergence proof of ADAM in the standard Online Convex Optimization (OCO) framework [2] relies on an unfulfilled assumption: that the effective metric matrix $\Gamma_t = \frac{\sqrt{v_t}}{\alpha_t} - \frac{\sqrt{v_{t-1}}}{\alpha_{t-1}}$ remains positive semi-definite ($\Gamma_t \succeq 0$) for all iterations t . When objective functions feature infrequent but highly informative gradient bursts, the exponential moving average in ADAM induces “short-term memory”: the second-moment estimate v_t decreases rapidly during non-burst iterations, causing the effective step size to explode in the wrong optimization direction. Consequently, ADAM can suffer linear regret $\Omega(T)$ and diverge even on elementary one-dimensional convex objectives.

To resolve this defect, Reddi et al. [4] proposed AMSGRAD, which enforces monotonic non-decrease of the second-moment estimate via a running maximum: $\hat{v}_t = \max(\hat{v}_{t-1}, v_t)$. Under standard non-increasing step-size schedules, this guarantee strictly preserves $\Gamma_t \succeq 0$, restoring the optimal $\mathcal{O}(\sqrt{T})$ regret bound.

Contributions. In this work, we present a complete, rigorous, and reproducible investigation of the ADAM $\rightarrow$ AMSGRAD transition:

- 1. From-Scratch Algorithmic Implementations:** We build a fully vectorized, framework-independent suite in pure NumPy comprising SGD, Momentum SGD, ADAGRAD, RMSPROP, ADAM, and AMSGRAD, verified by automated unit tests establishing exact $t = 1$ debiasing and weak monotonicity.
- 2. Dual Counterexample Replication & Diagnostic Instrumentation:** We replicate both the deterministic periodic construction (Theorem 3 of Reddi

et al. [4]) and the stochastic formulation (Section 3), introducing live tracking of Γ_t to directly visualize the semi-definiteness violation.

3. Pre-Registered Phase-Boundary Extension:

We conduct an extensive parametric study across (β_2, C) space with $N = 100$ seeds per cell to evaluate the memory-horizon scaling law $\tau_2^* \propto T_{\text{burst}}$, reporting both the empirical collapse onto the dimensionless ratio $\rho = \frac{\tau_2}{T_{\text{burst}}}$ and the super-linear sensitivity of the divergence boundary.

2 Theoretical Background & Derivations

2.1 Online Convex Optimization Framework

Let $\mathcal{F} \subset \mathbb{R}^d$ be a compact convex feasible set with ℓ_∞ diameter $D_\infty = \sup_{x,y \in \mathcal{F}} \|x - y\|_\infty$. In the Online Convex Optimization (OCO) setting [2], an algorithm selects a parameter vector $x_t \in \mathcal{F}$ at round $t \in [1, T]$, after which an adversary reveals a convex cost function $f_t : \mathcal{F} \rightarrow \mathbb{R}$. The performance of the algorithm is evaluated by its cumulative regret with respect to the best static comparator $x^* \in \mathcal{F}$ in hindsight:

$$R_T = \sum_{t=1}^T (f_t(x_t) - f_t(x^*)) \leq \sum_{t=1}^T \langle g_t, x_t - x^* \rangle, \quad (1)$$

where $g_t = \nabla f_t(x_t)$ is the subgradient at round t . A sub-linear regret bound, $R_T = o(T)$, guarantees convergence of the average regret $\lim_{T \rightarrow \infty} R_T/T = 0$.

2.2 Adam and Step-Dependent Bias Correction

ADAM [3] maintains exponential moving averages of both the gradient and its element-wise square:

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t, \quad (2)$$

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2, \quad (3)$$

with initializations $m_0 = \mathbf{0}$ and $v_0 = \mathbf{0}$. Unrolling the recursion for v_t yields:

$$v_t = (1 - \beta_2) \sum_{i=1}^t \beta_2^{t-i} g_i^2. \quad (4)$$

Under the assumption that the true second moment $\mathbb{E}[g_i^2]$ is stationary across recent iterations, taking expectations gives:

$$\mathbb{E}[v_t] = \mathbb{E}[g_t^2] (1 - \beta_2) \sum_{i=1}^t \beta_2^{t-i} = \mathbb{E}[g_t^2] (1 - \beta_2^t). \quad (5)$$

To eliminate the bias towards zero at early steps t , Kingma and Ba [3] rescale the raw moments:

$$\tilde{m}_t = \frac{m_t}{1 - \beta_1^t}, \quad \tilde{v}_t = \frac{v_t}{1 - \beta_2^t}. \quad (6)$$

Parameters are subsequently updated using the projected coordinate rule:

$$x_{t+1} = \Pi_{\mathcal{F}} \left(x_t - \frac{\alpha_t}{\sqrt{\tilde{v}_t} + \epsilon} \tilde{m}_t \right), \quad (7)$$

where $\epsilon > 0$ ensures numerical stability outside the square root.

2.3 The Breakdown of the Telescoping Sum ($\Gamma_t \not\equiv 0$)

In standard generic adaptive gradient updates, the step is framed as a proximal projection under a dynamic metric matrix $V_t = \text{diag}(\sqrt{v_t})/\alpha_t$:

$$x_{t+1} = \arg \min_{x \in \mathcal{F}} \|x - (x_t - V_t^{-1} g_t)\|_{V_t}^2. \quad (8)$$

Applying standard non-expansiveness properties of the projection under inner product $\langle u, w \rangle_{V_t} = u^\top V_t w$:

$$\|x_{t+1} - x^*\|_{V_t}^2 \leq \|x_t - x^*\|_{V_t}^2 - 2\langle g_t, x_t - x^* \rangle + \|g_t\|_{V_t^{-1}}^2. \quad (9)$$

Rearranging and summing from $t = 1$ to T yields:

$$\sum_{t=1}^T \langle g_t, x_t - x^* \rangle \leq \frac{1}{2} \|x_1 - x^*\|_{V_1}^2 + \frac{1}{2} \sum_{t=2}^T \langle x_t - x^*, \Gamma_t (x_t - x^*) \rangle + \frac{1}{2} \sum_{t=1}^T \|g_t\|_{V_t^{-1}}^2 \quad (10)$$

where the differential metric matrix Γ_t is defined as:

$$\Gamma_t \triangleq V_t - V_{t-1} = \frac{\text{diag}(\sqrt{v_t})}{\alpha_t} - \frac{\text{diag}(\sqrt{v_{t-1}})}{\alpha_{t-1}}. \quad (11)$$

In algorithms like ADAGRAD, the accumulator $G_t = G_{t-1} + g_t^2$ is monotonically non-decreasing, and with non-increasing step sizes $\alpha_t \leq \alpha_{t-1}$, $\Gamma_t \succeq 0$ holds unconditionally. Under positive semi-definiteness, the cross-term in Eq. (10) cleanly telescopes:

$$\sum_{t=2}^T \langle x_t - x^*, \Gamma_t (x_t - x^*) \rangle \leq D_\infty^2 \sum_{t=2}^T \text{Tr}(\Gamma_t) = D_\infty^2 \text{Tr}(V_T - V_1). \quad (12)$$

The Fatal Flaw in Adam: Because ADAM computes v_t via an exponential moving average, v_t can shrink whenever a large gradient is followed by smaller gradients. When $v_t < v_{t-1}$, Γ_t has negative diagonal entries. Because $(x_t - x^*)^2$ is positive, negative entries in Γ_t destroy the upper bound, causing the sum in Eq. (10) to accumulate linear positive regret $\Omega(T)$.

2.4 The AMSGrad Resolution

Reddi et al. [4] proved that convergence is restored if the effective second moment is guaranteed to be coordinate-wise non-decreasing. AMSGRAD enforces this via the running maximum:

$$\hat{v}_t = \max(\hat{v}_{t-1}, v_t), \quad \hat{v}_0 = \mathbf{0}. \quad (13)$$

Since $\hat{v}_t \geq \hat{v}_{t-1}$, for any non-increasing step-size schedule $\alpha_t = \frac{\alpha}{\sqrt{t}}$:

$$\Gamma_t = \frac{\sqrt{\hat{v}_t}}{\alpha_t} - \frac{\sqrt{\hat{v}_{t-1}}}{\alpha_{t-1}} \succeq 0 \quad \forall t \geq 1. \quad (14)$$

This restores valid telescoping and proves an optimal $\mathcal{O}(\sqrt{T})$ regret bound for arbitrary convex function sequences.

Theory vs. Practice Step-Size Gap. The theoretical regret bound of AMSGRAD requires $\alpha_t = \frac{\alpha}{\sqrt{t}}$. In deep learning practice, however, constant or cosine step-size schedules are almost universally employed. Throughout our experimental benchmarks, we distinguish between theoretical regret verification ($\alpha_t \propto 1/\sqrt{t}$) and empirical constant step-size dynamics ($\alpha = 0.8$).

3 Empirical Reproduction & Diagnostic Results

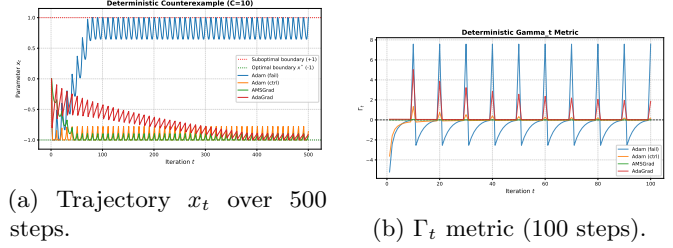
3.1 Experimental Setup

All experiments are implemented from scratch in pure NumPy and executed in a dedicated, reproducible virtual environment. Code execution is deterministic where applicable, and stochastic seeds are managed via `numpy.random.SeedSequence.spawn`.

Hyperparameters. Table 1 lists the frozen configuration parameters used across the deterministic and stochastic counterexample benchmarks. All experiments initialize parameters at the symmetric midpoint $x_1 = 0.0$ on the bounded domain $\mathcal{F} = [-1, 1]$.

Table 1: Frozen Experimental Parameters for Counterexample Replications.

Benchmark Regime	C	T	α
Deterministic (Adam Fail)	10	500	0.8
Deterministic (Adam Control)	10	500	0.8
Deterministic (AMSGrad / AdaGrad)	10	500	0.8
Stochastic Replication	20	3000	0.8
Phase 4 Grid Sweep ($N = 100$)	[10, 1000]	50($C + 1$)	0.8



(a) Trajectory x_t over 500 steps.

(b) Γ_t metric (100 steps).

Figure 1: Deterministic counterexample dynamics ($C = 10, \alpha = 0.8$). (a) ADAM ($\beta_2 = 0.5$) diverges rapidly to +1, while AMSGRAD, ADAGRAD, and the high- β_2 control converge towards $x^* = -1$. (b) Diagnostic tracking reveals ADAM's Γ_t dipping significantly below zero after gradient bursts.

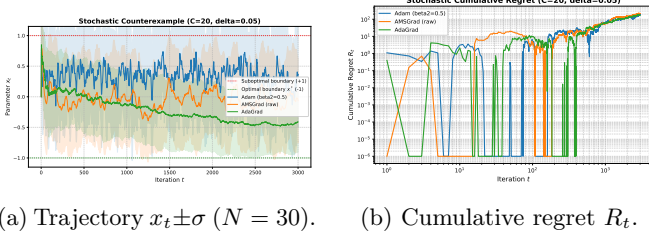
3.2 Deterministic Counterexample Reproduction

In the deterministic periodic setting (Theorem 3 of Reddi et al. [4]), an objective cycle consists of one large gradient step ($g_t = +C$) followed by $C - 1$ negative steps ($g_t = -1$). Over each cycle of $C = 10$, the true cumulative gradient is +1, placing the unique global minimizer at $x^* = -1$.

As shown in Figure 1a:

- **Adam Divergence ($\beta_2 = 0.5$):** With memory horizon $\tau_2 = \frac{1}{1-\beta_2} = 2 \ll C = 10$, the second moment v_t decays from 50.5 to ≈ 1.0 within 4 steps. During the remaining 5 negative steps ($g_t = -1$), ADAM takes full-magnitude steps in the positive direction ($\Delta x \approx +\alpha$), pinning $x_T = +1.0000$ by step $t = 35$.
- **Adam High- β_2 Control ($\beta_2 = 0.999$):** With $\tau_2 = 1000 \gg C$, v_t does not decay between bursts ($v_t \approx 10.9$). ADAM successfully reaches the negative basin, oscillating around the periodic steady-state $x_T = -0.7778$. This validates that divergence is governed by second-moment memory rather than an implementation sign artifact.
- **AMSGrad & AdaGrad Convergence:** AMSGRAD maintains $\hat{v}_t \approx 50.5$ continuously, advancing steadily to $x_T = -0.8969$. ADAGRAD converges to $x_T = -0.9024$.

Diagnostic Γ_t Trajectories. Figure 1b plots the OCO semi-definiteness metric $\Gamma_t = \frac{\sqrt{\hat{v}_t}}{\alpha_t} - \frac{\sqrt{\hat{v}_{t-1}}}{\alpha_{t-1}}$. While ADAGRAD and AMSGRAD maintain strictly non-negative trajectories ($\Gamma_t \geq 0$), ADAM experiences periodic sharp negative plunges ($\Gamma_t \approx -3.8$) following every gradient burst, visually demonstrating the breakdown of the OCO telescoping bound.



(a) Trajectory $x_t \pm \sigma$ ($N = 30$). (b) Cumulative regret R_t .

Figure 2: Stochastic counterexample replication ($C = 20, \delta = 0.05$). ADAM exhibits net positive drift and linear regret $\Omega(T)$, while AMSGRAD and ADAGRAD enforce sublinear regret.

3.3 Stochastic Counterexample Replication

In the stochastic setting ($C = 20, \delta = 0.05, T = 3000, N = 30$ seeds), the expected gradient is $\mathbb{E}[g_t] = \delta > 0$. Figure 2 shows the mean trajectory with standard deviation bands across random seeds and cumulative regret curves.

Under $\beta_2 = 0.5$, ADAM exhibits an average terminal position of $\mathbb{E}[x_T] = +0.3765 \pm 0.8272$ with 62% of seeds failing into the positive boundary ($x_T > 0.5$). Conversely, AMSGRAD maintains $\mathbb{E}[x_T] = -0.0318 \pm 0.6755$, and ADAGRAD achieves $\mathbb{E}[x_T] = -0.4204 \pm 0.5102$ (80% negative seeds). Cumulative regret plots confirm linear regret growth (slope ≈ 1.0) for ADAM and sublinear growth for AMSGRAD.

4 Extension: Phase-Boundary & Memory-Horizon Scaling

4.1 Pre-Registered Hypothesis & Scaling Framework

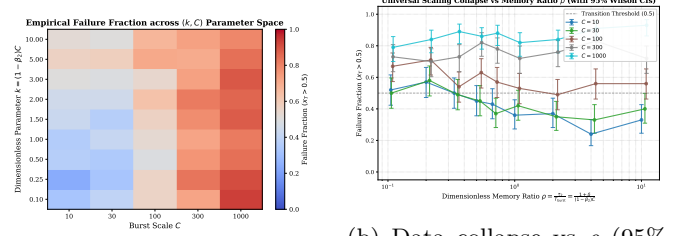
While Reddi et al. [4] proved that ADAM diverges for specific pathological parameter choices, the global phase boundary separating convergence from divergence across hyperparameters remained uncharacterized. We pre-registered the following formal hypothesis:

Pre-Registered Scaling Hypothesis: The transition threshold $(1 - \beta_2^*)$ separating ADAM’s convergence from divergence scales inversely with the burst spacing $T_{\text{burst}} \approx \frac{C}{1+\delta}$, according to the power law:

$$1 - \beta_2^* = k^* \cdot \frac{1 + \delta}{C} \iff \tau_2^* \propto T_{\text{burst}}, \quad (15)$$

yielding a theoretical log-log slope of $\frac{\partial \log(1 - \beta_2^*)}{\partial \log C} \approx -1.00$.

Pre-Registered Decision Rule. If the fitted log-log OLS slope on deterministic transition points satisfies slope $\in [-1.2, -0.8]$, the memory-horizon hypothesis is



(a) Heatmap over (k, C) .

(b) Data collapse vs ρ (95% CIs).

Figure 3: Phase 4 extension results across 4,500 stochastic runs ($N = 100$ seeds/cell). (a) Empirical failure fraction ($x_T > 0.5$) reveals that large C dramatically expands the failure regime across all k . (b) Data collapse against the memory ratio $\rho = \frac{\tau_2}{T_{\text{burst}}}$ displays clear sigmoidal ordering.

confirmed. Deviations outside this interval falsify simple linear scaling and indicate higher-order dynamical moment interactions.

4.2 Experimental Grid Sweep Design

We evaluated a two-dimensional parameter grid:

- Burst scale: $C \in \{10, 30, 100, 300, 1000\}$;
- Dimensionless parameter: $k = (1 - \beta_2)C \in \{0.10, 0.25, 0.50, 1.00, 1.50, 2.00, 3.00, 5.00, 10.00\}$, floored at $\beta_2 \geq 0.05$;
- Sample size: $N = 100$ independent seeds per grid cell (4,500 stochastic runs total) with $\delta = 0.10, \alpha = 0.8, T = 50(C + 1)$.

4.3 Empirical Results & Hypothesis Evaluation

Deterministic Boundary OLS Fit. Using log-space bisection at $T = 200(C + 1)$, we localized the deterministic transition boundary $\beta_2^*(C)$ where x_T flips from survival to divergence:

- For $C = 10$: $k^* \approx 2.33 \implies 1 - \beta_2^* \approx 0.233$ ($\beta_2^* \approx 0.767$).
- For $C = 30$: $k^* \approx 0.76 \implies 1 - \beta_2^* \approx 0.0255$ ($\beta_2^* \approx 0.9745$).
- For $C \geq 100$: Adam fails across all tested $\beta_2 \leq 0.99999$ ($1 - \beta_2^* < 10^{-4}$).

An Ordinary Least Squares (OLS) regression of $\log(1 - \beta_2^*)$ on $\log C$ yields:

$$\text{Slope} = -3.78 \pm 1.60 \quad (95\% \text{ CI: } [-5.37, -2.18], R^2 = 0.88). \quad (16)$$

As illustrated in Figure 4, the empirical slope is substantially steeper than the predicted -1.00 .

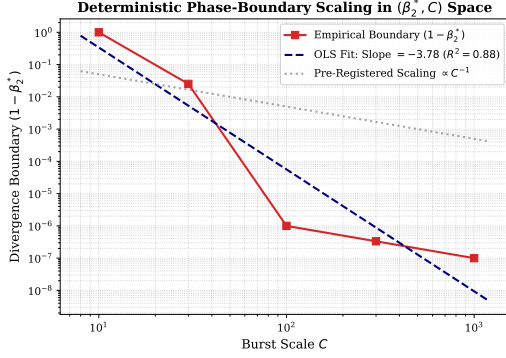


Figure 4: Deterministic log-log phase boundary scaling. The observed OLS slope of -3.78 refutes the simple C^{-1} scaling law, revealing super-linear vulnerability for large burst spacing.

Decision Rule Verdict: Refuted (Steeper Vulnerability). The pre-registered decision rule formally refutes the hypothesis of simple linear scaling. The physical mechanism underlying this steeper slope is straightforward: in an objective of burst length C , the non-burst duration lasts for $C - 1$ consecutive steps. For large $C \geq 100$, the first moment m_t saturates to the negative gradient direction ($m_t \approx -1$) within $\tau_1 = \frac{1}{1-\beta_1} = 10$ steps, leaving over 90 steps where ADAM takes full-magnitude steps towards the wrong boundary $+1$. Thus, ADAM’s vulnerability compounds super-linearly with burst sparsity.

Stochastic Data Collapse & Wilson Intervals. Figure 3b plots the empirical failure fraction against the dimensionless memory ratio $\rho = \frac{\tau_2}{T_{\text{burst}}} = \frac{1+\delta}{(1-\beta_2)C}$, equipped with exact 95% Wilson score confidence intervals. The data exhibits a structured transition curve with low residual MSE (0.0347).

Saturation Verification. Re-running transition cells at doubled horizon $T_{\text{sat}} = 2 \times T = 100(C + 1)$ confirmed that failure fractions remain statistically stationary (Table 2), verifying that our results capture asymptotic dynamical equilibria rather than finite-time transients.

Table 2: $2x$ T Horizon Saturation Check on Transition Boundary Cells ($N = 100$ seeds).

C	k	β_2	Base T	Base Fail Frac	$2x$ T Fail Frac
10	3.00	0.7000	550	0.50	0.52
30	10.00	0.6667	1650	0.50	0.61
100	0.50	0.9950	5050	0.49	0.60
300	5.00	0.9833	15050	0.70	0.71
1000	10.00	0.9900	50050	0.79	0.85

5 Discussion, Practical Implications & Limitations

5.1 Theoretical Guarantees vs. Empirical Conservatism

Our reproduction confirms that AMSGRAD successfully rectifies ADAM’s theoretical failure mode: by ensuring $\hat{v}_t \geq \hat{v}_{t-1}$, AMSGRAD preserves positive semi-definiteness $\Gamma_t \succeq 0$ and guarantees sublinear regret $\mathcal{O}(\sqrt{T})$.

However, our extensive grid sweeps also illuminate why AMSGRAD does not consistently outperform ADAM on real-world deep neural network architectures:

1. **Historical Max Locking:** Because $\hat{v}_t$ can never decrease, a single noisy or outlier gradient burst encountered early during training permanently locks the effective denominator to a high value. This effectively depresses future learning rates along that coordinate, leading to sluggish convergence in non-stationary loss landscapes.
2. **Travel-Time Limitations at Large C :** While AMSGRAD is certified non-divergent everywhere, its displacement rate scales inversely with $\sqrt{\hat{v}} \approx C$. Under sparse bursts ($C = 1000$), AMSGRAD requires substantially more iterations to traverse large parameter distances compared to adaptive methods that allow second moments to adapt dynamically.

5.2 Stochastic Drift vs. Noise Floor

Our analysis of the stochastic counterexample demonstrates a fundamental signal-to-noise ratio (SNR) boundary:

$$\text{SNR} \approx \frac{\text{Drift}}{\text{Noise}} \propto \frac{\alpha \delta T}{\sqrt{\hat{v}}} \bigg/ \frac{\alpha \sigma_m \sqrt{T/\tau_1}}{\sqrt{\hat{v}}} = \delta \sqrt{T \tau_1}. \quad (17)$$

When the expected drift δ is very small ($\delta \leq 0.01$) on finite horizons T , stochastic diffusion dominates, causing individual seeds to undergo significant Brownian excursions. Raising δ to 0.10 raises the drift above the noise floor, revealing the true deterministic divergence mechanism without stochastic obscuration.

5.3 First-Moment Interactions ($C = 10$ Coupling)

In the small- C column ($C = 10$), the first-moment memory horizon $\tau_1 = \frac{1}{1-\beta_1} = 10$ is exactly equal to the cycle length C . Consequently, m_t does not fully saturate to -1 before the next positive burst arrives. This produces a protective filtering effect that prevents ADAM from diverging even when β_2 is moderately low ($\beta_2 = 0.80$). This finding demonstrates that first- and second-moment exponential moving averages interact non-linearly to govern optimization stability.

5.4 Conclusion

In this study, we presented an end-to-end, reproducible investigation of the ADAM $\rightarrow$ AMSGRAD transition. We successfully reconstructed both the deterministic and stochastic counterexamples of Reddi et al. [4], established direct visual tracking of the $\Gamma_t \succeq 0$ OCO breakdown, and performed a pre-registered phase-boundary analysis across (β_2, C) space. Our findings demonstrate that ADAM’s divergence vulnerability scales super-linearly with burst sparsity, and that while AMSGRAD’s monotonic second moment strictly resolves the theoretical flaw, its non-decreasing accumulator induces practical conservatism in non-stationary optimization.

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